**Video Transcript - Part 18**

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Hey everyone, welcome to part eighteen. In this video we're taking a closer look at three of the most important classification metrics: precision, recall, and the F1 score. We touched on them with the confusion matrix, so now we'll build real intuition for what each one means, how they trade off against each other, and when to use which.

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Here's the plan. We'll set up a small example, then go through precision, recall, and the tradeoff between them. After that we'll cover the F1 score and its cousin, F beta, see how these metrics work with more than two classes, and finish with how SageMaker lets you tune for them.

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Let's use a spam filter, where spam is the positive class. Imagine two hundred emails. Forty are actually spam, and one hundred sixty are not. The filter flags fifty emails as spam. Thirty of those really are spam, our true positives. Twenty are legitimate emails flagged by mistake, the false positives. It missed ten spam emails, the false negatives, and correctly passed one hundred forty good emails, the true negatives.

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Precision answers this question: when the model says spam, how often is it right? The formula is true positives divided by true positives plus false positives. In other words, true positives divided by everything the model flagged. Here that's thirty out of fifty, or sixty percent. Precision cares about false positives. If a filter has low precision, it's burying real emails in the spam folder. High precision means that when the model raises a flag, you can trust it.

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Recall asks a different question: of all the real spam, how much did the model find? The formula is true positives divided by true positives plus false negatives, or true positives divided by all actual positives. Here that's thirty out of forty, or seventy five percent. Recall cares about false negatives, the misses. Low recall means spam keeps slipping into the inbox. It's also called sensitivity or the true positive rate.

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The catch is that precision and recall pull against each other. Imagine the filter flags only the one email it's most sure about. If it's right, precision is a perfect one hundred percent, but recall is just two and a half percent, since it found one out of forty. Now imagine it flags every email as spam. Recall is one hundred percent, because it caught all the spam, but precision falls to twenty percent, since only forty of two hundred are spam. Your classification threshold slides you between these two extremes, so you always have to choose a balance.

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That's where the F1 score comes in. It combines precision and recall into one number, using the harmonic mean. The formula is two times precision times recall, divided by precision plus recall. For our filter, that's two times point six times point seven five, divided by one point three five, which is about sixty seven percent. Why not just average them? Because a simple average can hide a problem. Say precision is one hundred percent but recall is two percent. The average is fifty one percent, which looks okay, but the F1 score is only about four percent. F1 is high only when both precision and recall are high.

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Sometimes one matters more than the other, and that's what F beta is for. It's a generalization of F1 with a parameter, beta, that sets how much more you value recall than precision. With beta equal to one, you get F1, a balanced score. A larger beta, like two, weights recall more, which suits things like disease screening or safety checks where misses are costly. A smaller beta, like point five, weights precision more, which suits things like spam filtering or recommendations where false alarms are costly.

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Everything so far was for two classes. With more than two classes, you calculate precision, recall, and F1 for each class, and then combine them. Macro averaging takes the plain average across classes, so every class counts equally, which is great for seeing how you do on rare classes. Weighted averaging also averages across classes, but weights each by how many examples it has. And micro averaging pools all the predictions together first, then calculates a single score. If your classes are imbalanced, always look at macro or per class results, not just overall numbers.

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In SageMaker, you can tune directly for these metrics. The built in Linear Learner algorithm lets you choose how the best model is selected, using accuracy, F beta, precision at a target recall, or recall at a target precision. That last pair is very practical. You might say, I need at least ninety percent recall, now give me the highest precision possible. Automatic model tuning can also use validation precision, recall, or F beta as the objective metric. And remember to pair these metrics with the techniques for imbalanced data, like resampling and class weights.

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So to recap, precision tells you how trustworthy the positive predictions are, and recall tells you how many of the real positives you found. They trade off against each other as you move the threshold. F1 balances the two with a harmonic mean, and F beta lets you favor one or the other. Choose based on the cost of your mistakes, and look at per class results when you have more than two classes. Thanks for watching, and I'll see you in the next one.